



**ELECTRIC VEHICLE SIMULATION USING
MATHLAB/SIMULINK**

RUPESHWARAN PARANTHAMAN

STUDENT ID:23069976

1.INTRODUCTION.

The automotive industry is quickly moving toward electrification, and electric vehicles (EVs) are now seen as a key way to cut greenhouse gas emissions and make transportation more energy-efficient.[1] The electric driving range is one of the most important performance measures for an electric vehicle. It has a big impact on how well customers accept and use the vehicle. Manufacturers often advertise impressive range figures, but these numbers are usually based on standardized driving cycles like the Worldwide Harmonised Light Vehicles Test Procedure (WLTP)[4], which may not be a good way to show how cars actually drive.

The range of an electric vehicle is affected by a number of factors that are all related to each other. These include the battery's capacity, the vehicle's weight, the driving cycle, the ability to use regenerative braking, and the driver's behavior. For instance, urban stop-and-go conditions may allow for more energy recovery through regenerative braking, while driving at high speeds for a long time can greatly increase power demand and shorten range[1,3]. So, it's important to know how driving conditions affect power demand, energy use, and overall range in order to improve EV design and get a realistic picture of how well they work.

The goal of this project is to use MATLAB and Simulink to model and simulate a hypothetical electric vehicle, with a focus on looking at how far it can go on a single charge under different driving cycles. The study examines the variation in power demand relative to driving conditions, the consumption and regeneration of battery energy, and the impact of changes in regenerative braking efficiency on overall driving range. The project shows the flaws in standardized test cycles and gives us a better idea of how electric vehicles (EVs) work in the real world by comparing simulation results with the advertised range of a Tesla Model 3 under WLTP conditions.

2.PART A MODEL UPDATE AND SETUP.

QUESTION A1 MATLAB MODEL UPDATE:

The only changes made to the provided MATLAB script were to the two parameters listed in the "Provided Data" section: the battery nominal capacity (Ah) and the vehicle mass (kg). As instructed, these numbers were customized based on the last three digits of the Student Registration Number (SRN).

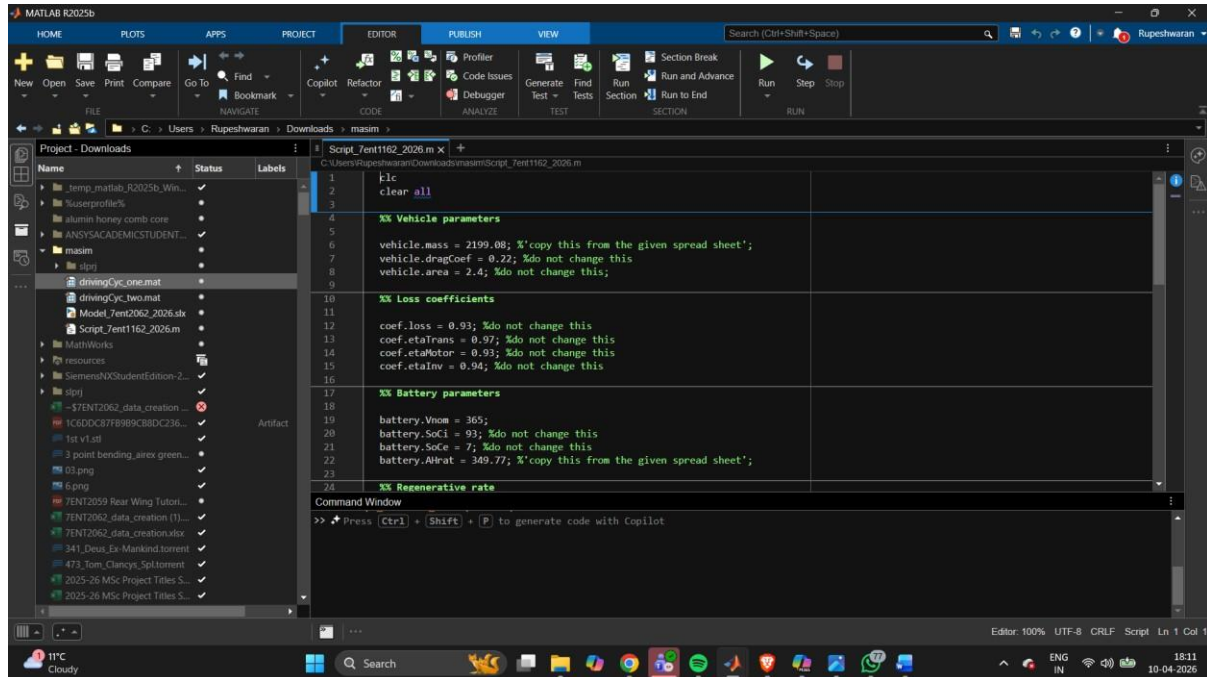
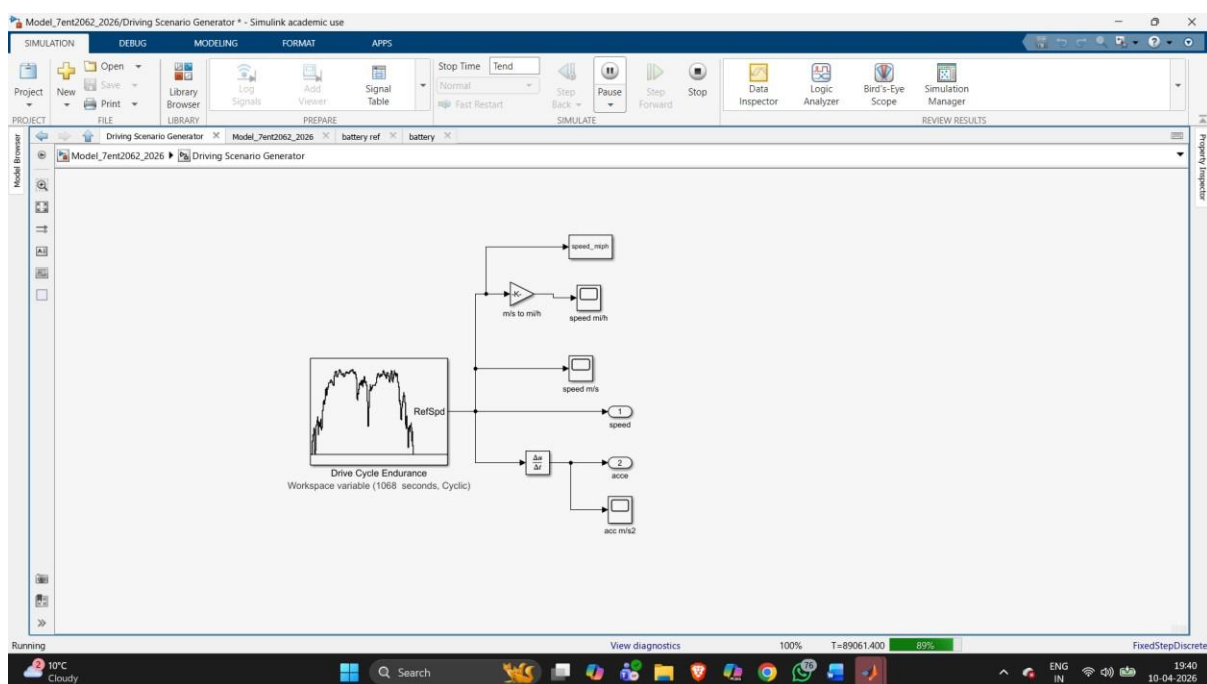
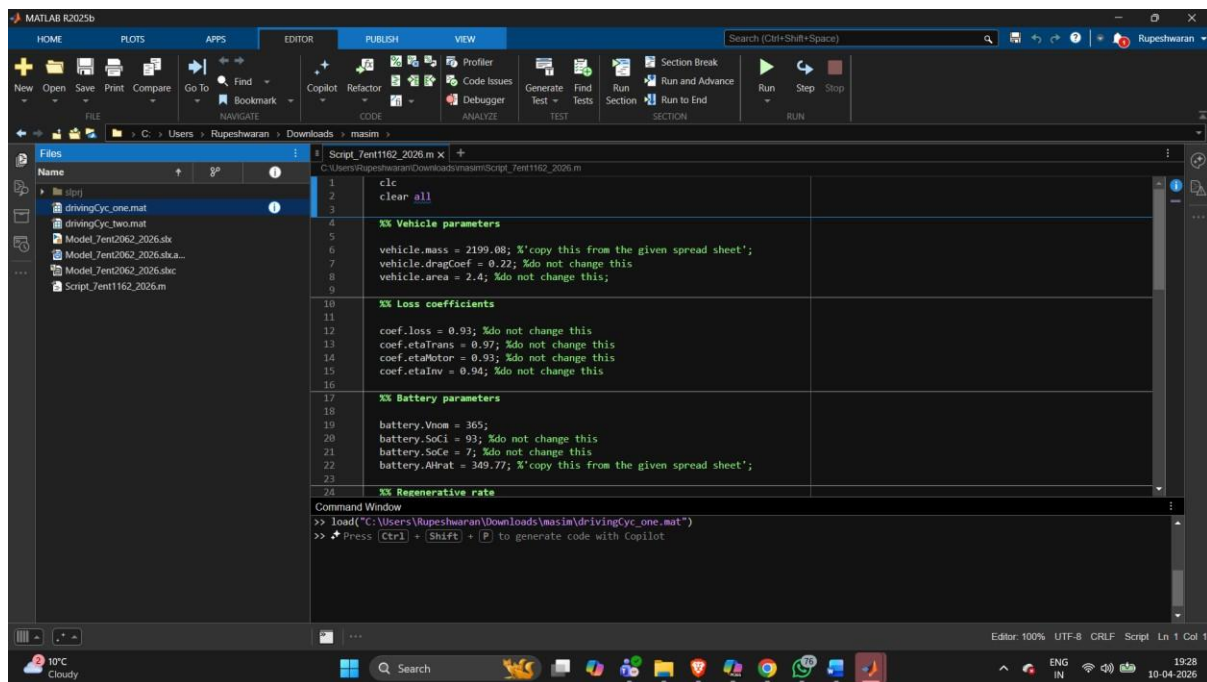


FIG 1.0 VALUE CHANGE.

All the downloaded files are saved under one folder named: MASIM. and the vehicle mass and battery AHrat (ampere-hour) line 6 and 22 are changed to 2199.08 and 349.77 according to my last three digits of my student registration number (SRN).

QUESTION A2: RUNNING CYCLE AND SIMULINK SETUP.

To set up the Simulink model, all of the files that were given, such as the MATLAB script, Simulink model, two driving cycles, and the spreadsheet, were put in the same working directory. The cd command in MATLAB set the directory, and the MATLAB script ran the simulation. Driving Cycle 1 was chosen and loaded. The script then automatically sent the driving cycle data and updated the vehicle parameters to the Simulink model. This made sure that the model was called directly from the script and ran without any problems with the chosen driving cycle. The Driving Scenario Generator subsystem, which is shown in the figure below, shows that Driving Cycle 1 worked as planned.



Driving Cycle 1 was chosen and loaded. The script then automatically sent the driving cycle data and updated the vehicle parameters to the Simulink model. This made sure that the model was called directly from the script and ran without any problems with the chosen driving cycle. The Driving Scenario Generator subsystem, which is shown in the figure above, shows that Driving Cycle 1 worked as planned.

3.PART B POWER DEMAND TRAJECTORY.

QUESTION B1: POWER DEMAND PLOTS.



FIG 3.1 POWER PLOT FOR DC1.

The graph presents the variation in electric power consumption (kW) with respect to time (s) for one iteration of Driving Cycle 1. It can be seen that there are several positive spikes in the power plot that signify the process of acceleration, as well as a few dips in the power plot that represent the process of regenerative braking.

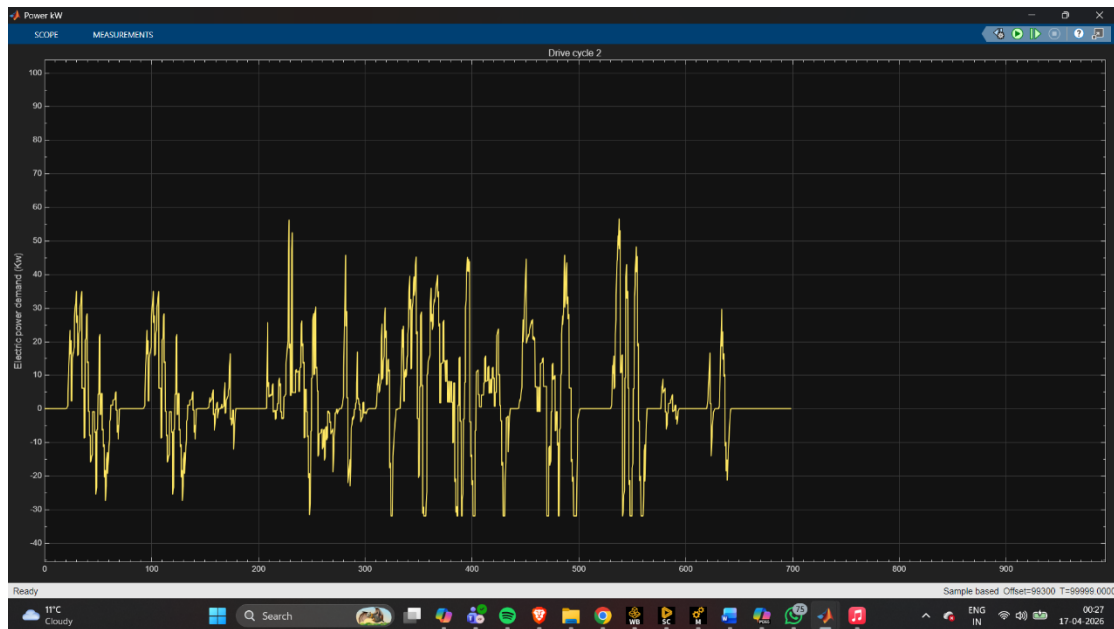


FIG 3.2 POWER PLOT FOR DC2

The graph shows the electrical power demand against time for Driving Cycle 2. The profile is smoother than the one in Driving Cycle 1 with longer intervals where power stays positive, while there are fewer sections where the power is negative. This suggests more stable vehicle movement with less braking activity, opportunities for regenerative braking and lower transient power demands than in an urban driving cycle.

QUESTION B2: EXPLANATION ON PEAKS AND PATTERNS IN THE POWER DEMAND TRAJECTORY.

Electrical Power Demand Trajectories can be discussed based on the principles of longitudinal dynamics of the vehicle. In order to accelerate, it is necessary to provide sufficient power to overcome the vehicle's mass and all the types of resistances, i.e. aerodynamic drag and rolling resistance[1]. Positive values in power demand curves will therefore appear during acceleration. The larger the acceleration is, the higher is the inertial force. It can be seen that during deceleration, there is a reduction in power demand; thus, the electric motor operates in generator mode, and it is possible to store the recovered energy in the battery[3]. Such a situation is reflected by the presence of negative values in the power demand profile.

In Driving Cycle 1, we can notice considerable fluctuations of the values in the power demand trajectory. There are many sharp power peaks and a lot of areas with negative power demand. That happens because of the frequent acceleration and braking events. Therefore, the power demands become high due to increased requirements of the transient tractive force, whereas more braking events mean more chances to recover energy from the electric motor. Driving Cycle 1 may thus be considered as an urban driving cycle.[3]

However, Driving Cycle 2 demonstrates less sudden variations in power demand and prolonged periods of steady power consumption. This suggests a consistent vehicular operation without frequent acceleration and braking maneuvers. Thus, air drag and rolling resistance contribute significantly to power demand in this scenario, rather than inertia. Consequently, there are fewer braking instances, leading to less regenerative braking. Such conditions apply to highway driving and extra-urban driving scenarios[1].

To determine the electrical energy for each of the driving cycles, the electrical power demand signal was integrated using the Integrator (1/s) block in the regeneration section of the Simulink model. The electrical energy is calculated as follows:

$$E = \int_0^T P(t) dt$$

where $P(t)$ is the electrical power requirement in kilowatts and T is the time required for one complete driving cycle in seconds. The output of the integrator gives cumulative energy in $\text{kW}\cdot\text{s}$. At the end of one driving cycle, the final value of the energy versus time waveform indicates the total energy expended during one driving cycle.

From the simulation data, the final value of the integrator for Driving Cycle 1 was $650 \text{ kW}\cdot\text{s}$, while for Driving Cycle 2 it was $640 \text{ kW}\cdot\text{s}$. This value was then converted into kilowatt hours by:

$$E_{\text{kWh}} = 3600 E_{\text{kW}\cdot\text{s}}$$

Therefore, electrical energy utilization by the motor is found to be **0.181 kWh** during Driving Cycle 1, whereas it is **0.178 kWh** during Driving Cycle 2. Regenerative braking energy being larger during Driving Cycle 1 indicates that it belongs to an urban type of drive cycle, while Driving Cycle 2 belongs to a highway.

4.PART C ENERGY CONSUMPTION AND ELECTRIC RANGE ANALYSIS.

QUESTION C1: ENERGY CONSUMPTION AND ELECTRIC RANGE ANALYSIS .

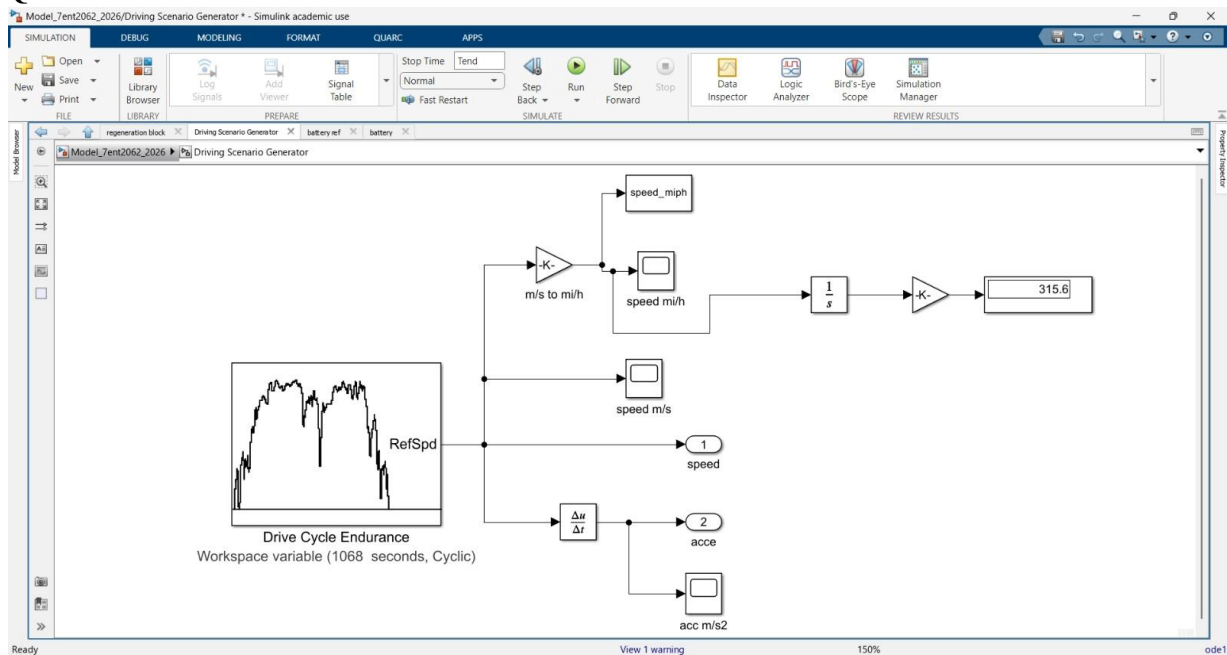


FIG 4.1 REGEN COEFF OF DC1.

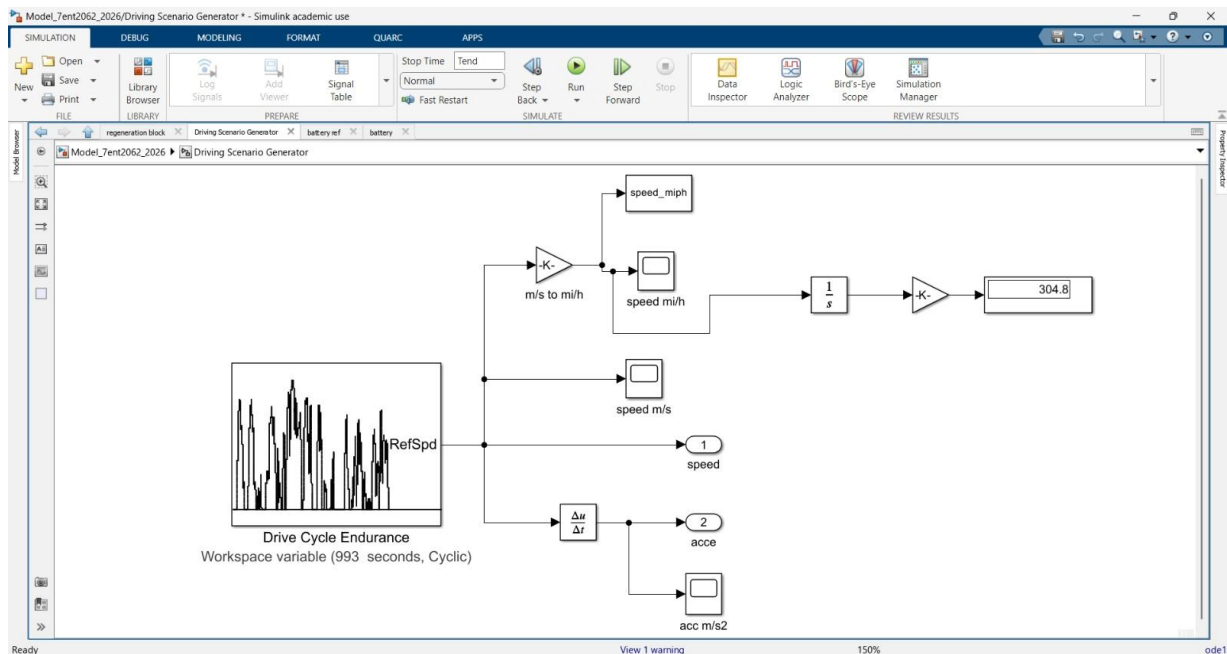


FIG 4.2 REGEN COEFF OF DC2.

The range of the electric vehicle for the speed profile in miles/hour was computed through integration in the Simulink simulation using the 1/s integrator block. The electric range of Drive Cycle 2 was determined to be 49.1 miles. The increased range means that Drive Cycle 2 is more efficient compared to a stop-and-go drive cycle, since it has less speed variation, less abrupt acceleration, and braking actions[1], resulting in reduced energy losses and an increased range.

The available battery capacity was computed based on the nominal battery voltage, battery capacity in ampere hours, and available state of charge (SOC)[2]:

Using:

- $V_{nom} = 365 \text{ V}$
- $AH_{rat} = 349.77 \text{ Ah}$
- $SoC_i = 93\%$
- $SoC_e = 7\%$

$$E_{usable} = \frac{365 \times 349.77 \times (93 - 42.89)}{1000}$$

$$E_{usable} = 63.97 \text{ kWh}/304.8 = 0.210 \text{ kWh/mile}$$

The energy consumption in kWh/mile is then:

$$EC = \frac{109.8}{0.210} = 522.8 \text{ kWh/mile}$$

So, for C1, Drive Cycle 2 gives:

- Range = 49.1 miles
- Usable battery energy = 109.8 kWh
- Energy consumption = 522.8 kWh/mile

Drive Cycle 2 has more efficient operations with increased range. It does so because its driving cycle has a better operation than that of Drive Cycle 1, since it uses less power and experiences fewer transients and thus less energy losses. Therefore, it consumes less fuel per mile, hence making it have more range.

QUESTION C2: RE-RUNING THE SIMULATION WITH MODIFIED REGENERATION COEFFICIENT.

<pre>%% Battery parameters battery.Vnom = 365; battery.SoCi = 93; %do not change this battery.SoCe = 7; %do not change this battery.AHrat = 349.77; %'copy this from the given spread sheet';</pre>	
<pre>%% Regenerative rate regenRate = 0.65; % this factor will change to 0.65 in QC2</pre>	
<pre>%% Driving cycle data load drivingCyc_one.mat % two driving cycles will be compared, change will be required in QB/C, load accordingly</pre>	

FIG 4.3 MODIFIED REGENERATION COEFFICIENT.

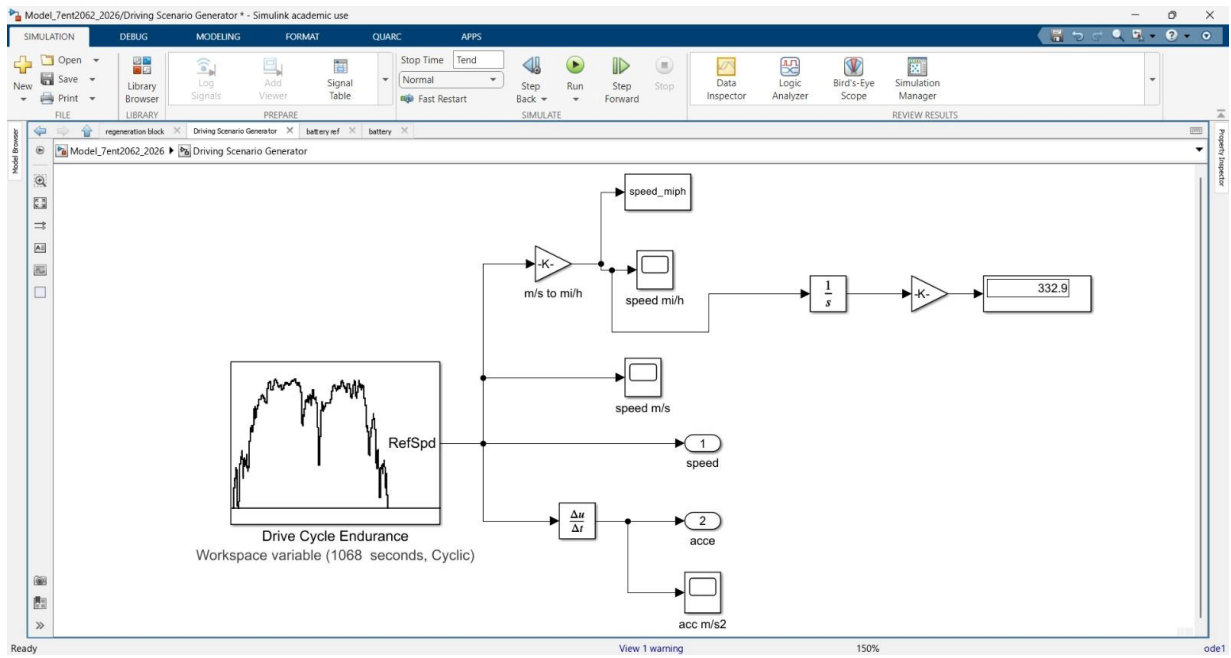


FIG 4.4 REGEN COEFF OF DC1.

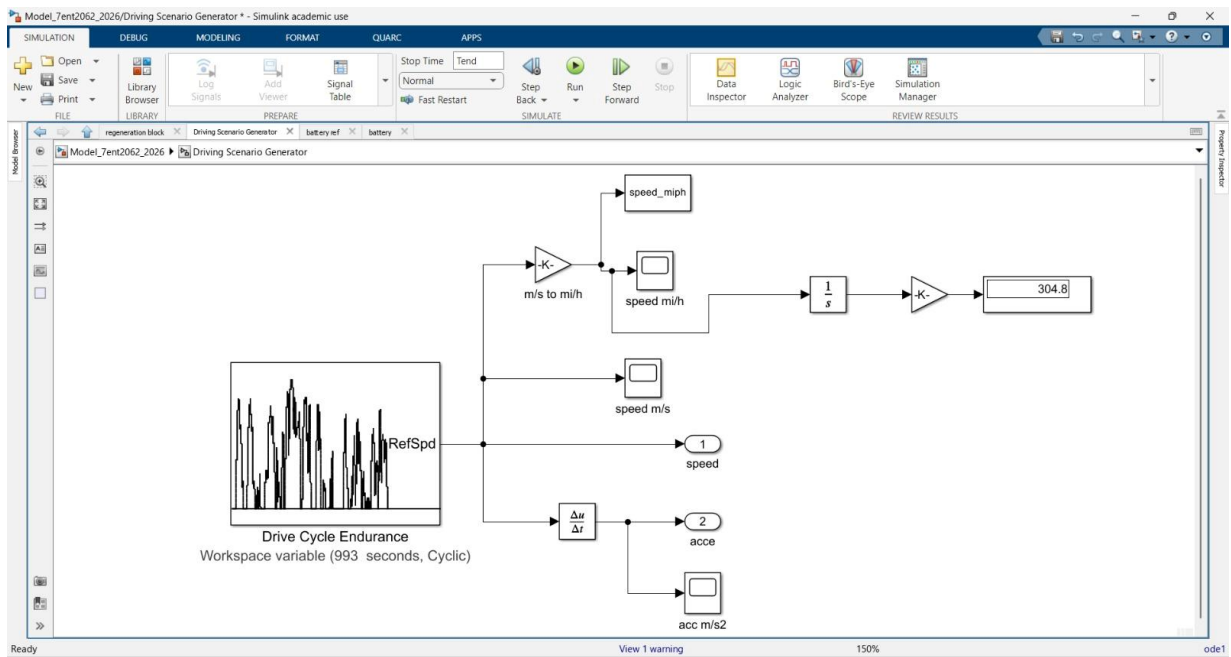


FIG 4.5 REGEN COEFF OF DC2.

In case of C2, the regen Rate was changed from 0.25 to 0.65. For Drive Cycle 2, the electric range was similar but the final state-of-charge increased from 42.89% to 49.51%. It means that the car generated more energy when decelerating and consumed less battery power.

$$E_{usable} = \frac{365 \times 349.77 \times 0.93 - 49.51}{1000} = 118.68 \text{ kWh}$$

$$E_{usable} = 118.68 \text{ kWh} / 304.8 = 0.389 \text{ kWh/mile}$$

The energy consumption in kWh/mile is then:

$$EC = \frac{109.8}{0.389} = 282.3 \text{ kWh}$$

- Range = 49.1 miles
- Usable battery energy = 109.8 kWh
- Energy consumption = 282.3 kWh/mile

That is, Drive Cycle 2 has better efficiency and range. Practically speaking, an increase in regen Rate implies an increased capability of the vehicle to utilize the braking energy[2,3]. This is affected by many factors, including the capability of the motor-generator system, inverter, battery management system, state of charge of the battery, and battery temperature[2]. Also affected is the driving profile, because there will be more opportunities for regeneration when compared with highway driving.

Regenerative braking technology is a big step towards increasing efficiency, particularly when used in urban areas[1,3], but it cannot totally substitute for the available battery energy. If anything, it could assist in reducing the battery capacity required while maintaining a sufficient range, though up to a certain extent. This is due to the fact that the reduced battery capacity will continue to be limited by other factors such as cruising demands, acceleration demands, and the period with little braking involved.

C2.1: REGENERATIVE BRAKING IMPACT AND BATTERY DOWNSIZE TRADING-OFF.

Regenerative braking is an important efficiency factor for EVs, especially in scenarios where vehicles stop frequently. From the simulation findings, the increase in the value of regeneration from 0.25 to 0.65 allowed the system to capture a higher percentage of energy during braking and minimized battery drain. Specifically, in Drive Cycle 2, the final state of charge rose from 42.89% to 49.51%. This implies that there was a higher amount of braking energy harnessed and used to charge the battery. Thus, the vehicle required less energy stored in the battery to traverse the driving cycle, despite the fact that the distance remained unchanged.

This phenomenon is especially visible in an urban environment because urban roads are characterized by numerous acceleration and decelerations. Each time the car slows down, some portion of its kinetic energy can be recycled instead of dissipating into the brake pads as heat. The regen Rate parameter increases the portion of energy that is absorbed by the batteries. Nevertheless, the energy itself remains dependent on the strength of braking, the car's speed, and how long the slowing-down process takes. In a smooth cycle with reduced braking, there will be fewer occasions for the absorption of energy, and thus, the effect will be minimized.

In fact, regen Rate tuning is not only a function of driving cycle. It is influenced by the capabilities of the vehicle components as well as the battery pack status. The motor-generator needs to be efficient at generating power, while the inverter needs to be able to manage the recovered electrical power, and the battery needs to be charged in such a way that the current will be handled effectively. In addition, the battery's SoC is critical; it cannot accept much regenerated energy when the battery is fully charged. Likewise, the battery temperature is also a factor; when the battery is cold, it can accept little charging capacity.

Regarding battery down-sizing as another benefit, regenerative braking can help with that as well; however, its role here will be partial only. For example, the more energy the vehicle manages to recover when braking, the less energy will need to be provided from the battery to complete the same distance. In that case, a battery pack of reduced size could be considered sufficient without compromising its range.

Nevertheless, there are limits to how much the battery requirements can be decreased in this way. The battery will still have to provide energy for acceleration, moving at constant speeds, driving up inclines, and situations where no brake application is performed. Therefore, regenerative braking will not be able to take the place of a sufficient battery. Instead, regenerative braking can make the usage of a battery more efficient.

To sum it up, regenerative braking turns out to be a very important aspect of improving the efficiency and reducing the energy lost by electric vehicles, especially in city conditions. At the same time, regenerative braking cannot completely replace the functions of battery packs[5].

REFERENCES

1. **Ehsani M, Gao Y, Longo S, Ebrahimi K.**
Modern electric, hybrid electric, and fuel cell vehicles. 3rd ed. Boca Raton (FL): CRC Press; 2018.
2. **Husain I.**
Electric and hybrid vehicles: design fundamentals. 2nd ed. Boca Raton (FL): CRC Press; 2011.
3. **Gao Y, Chen L, Ehsani M.**
Investigation of the effectiveness of regenerative braking for electric and hybrid electric vehicles. *SAE Technical Paper*. 1999;1999-01-2910.
4. **United Nations Economic Commission for Europe (UNECE).**
Worldwide harmonised light vehicles test procedure (WLTP). Geneva: UNECE; 2017.
5. **Chan CC.**
The state of the art of electric and hybrid vehicles. *Proceedings of the IEEE*. 2002;90(2):247-275.